Assignment 1

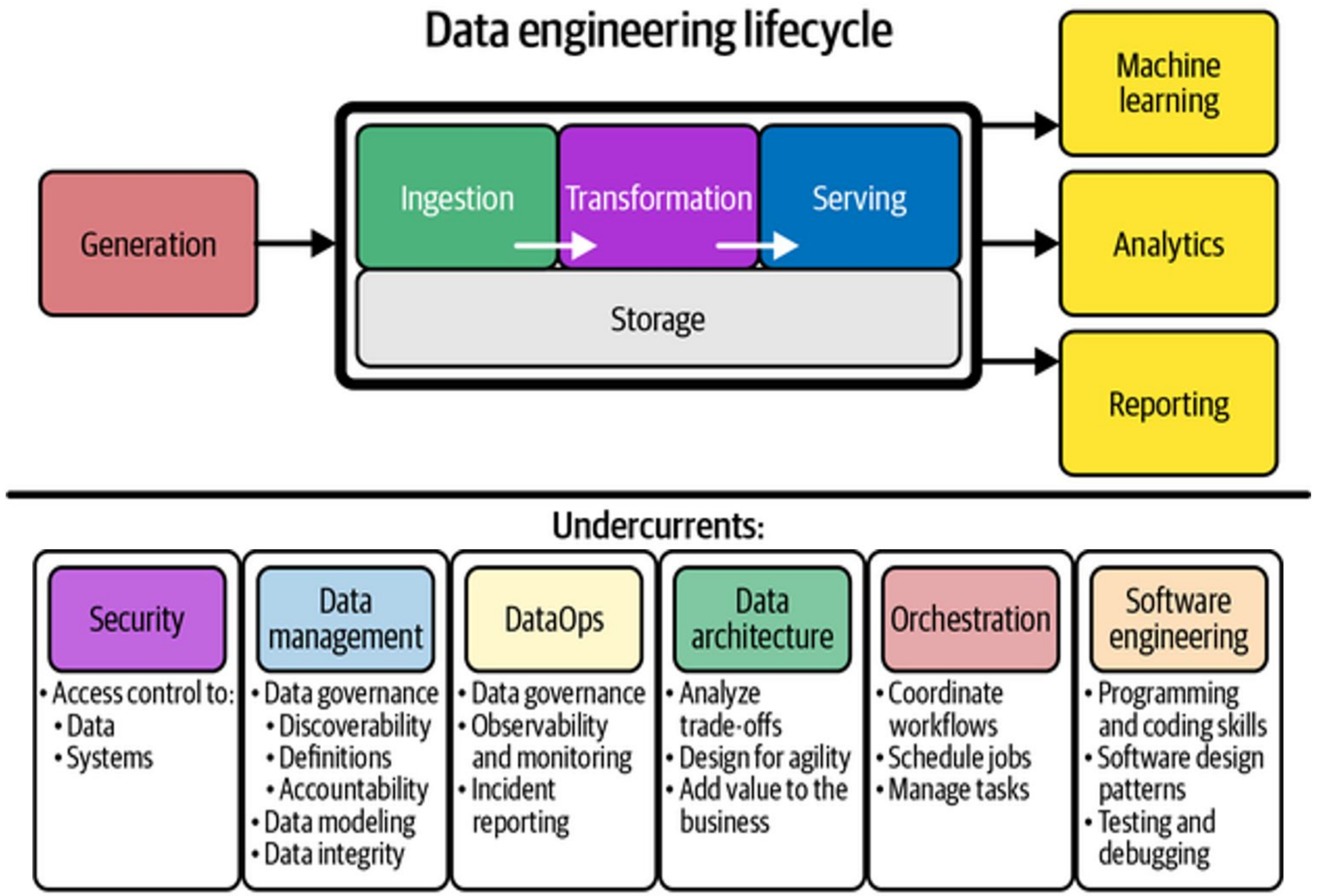
INDEX

Context

Context

Assignments in Big Data Technologies pursue a holistic understanding and management of the data lifecycle in general and the data engineering lifecycle in particular.

The student will face a situation where generating value from data is not obvious. Leveraging big data technologies will be the way to manage the questions that will come up before serving intelligence with trust.

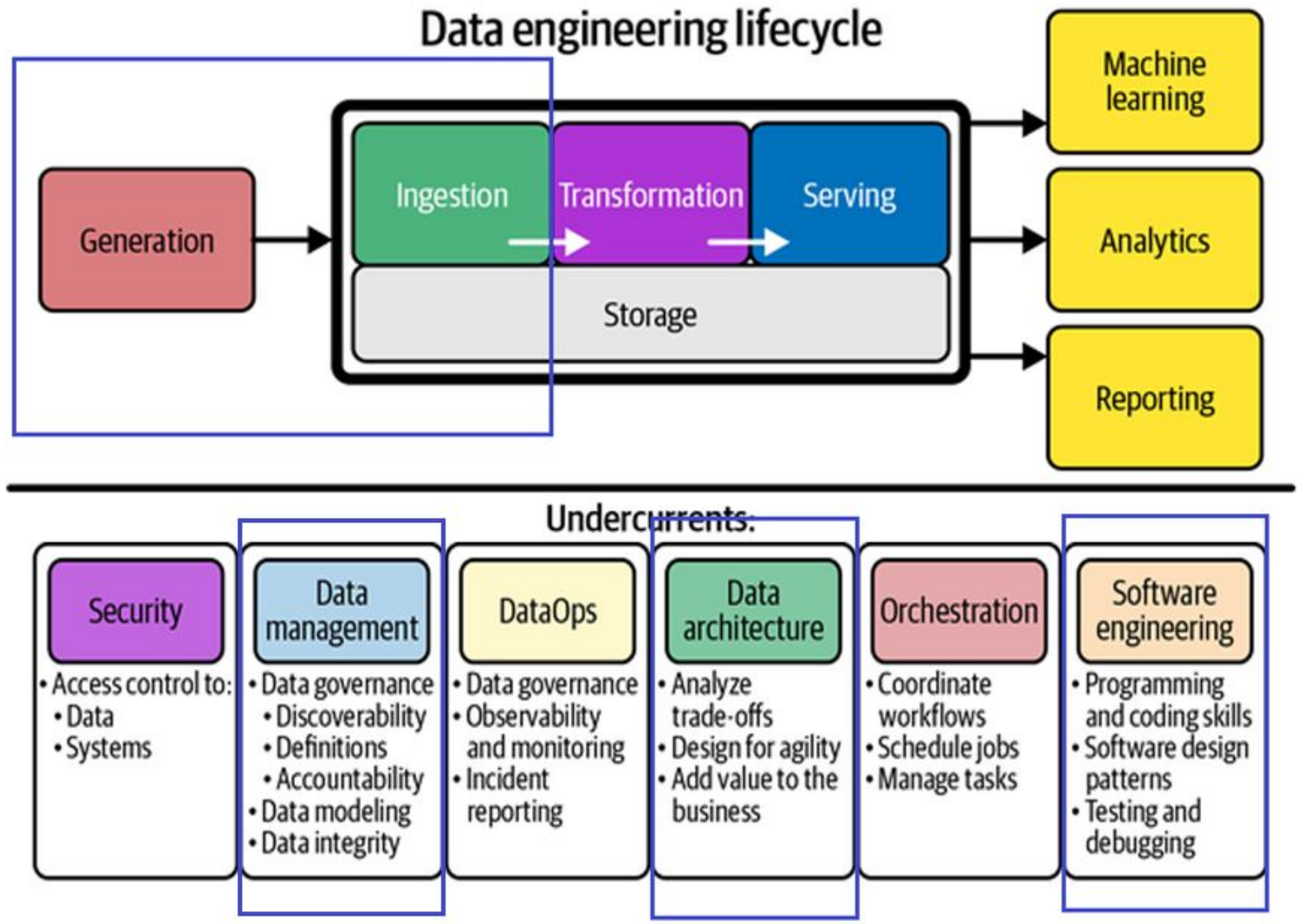


Assignment 1

Data integration

The Assignment 1 focuses on the left side of the data lifecycle: generation and ingestion of data. When we combine this two tasks the result is a **data integration** pipeline.

Besides, the student will have to design the data architecture chosen to solve the problem, the software components that will incorporate to the solution and the data dictionary that describes the data.



Statement

Can we generate value moving data from point A to point B? Data integration is considered one of the most challenging steps of the data lifecycle due to the variety of the new data producers and systems emerging every day. Data is continuously produced and delivering it from the capture to the storage system reliably becomes a tough task.

A prerequisite to integrate data is a deep understanding of the structure of the data, its schema, its data types, the constraints around them and the surrounding systems that will support a reliable data movement. Once the data lands in the storage layer it should be ready to flow downstream.

This assessment aims to validate that the student can design and implement a reliable data integration pipeline and test its behavior once it is working.

Statement

Generation and data management

The step of data generation requires the student to produce a dataset. The student can use open data sources to build the set of assets from a desired topic of free choice. The dataset will include, but is not limited to:

- One csv file
- One json file
- Directly, programmatically-generated datasets in SQL

Once the student has gathered (or generated) the files, the student will analyze the metadata of the dataset. Next to the dataset, the student will generate a data dictionary that details what is inside the dataset.

Data architecture design

The student will design a diagram of the architecture that will enable the data integration. The systems, tools or services are of free choice. Open source is recommended but not mandatory.

In addition to the diagram, the student will explain the reason behind the selection of each component and the benefits of using them regarding reliability, scalability and maintainability of the entire system and the resources available alongside the identification of possible roles associated with each involved process.

Statement

Data ingestion pipeline and software engineering
The data ingestion of the dataset to a storage system will have an extraction step from the files and a load step into the database(s) desired. In case the dataset (or directly the database) are entirely generated programmatically, the student will provide the generation code, whether python, SQL, vbNET or any other thereby used code snippets/scripts/software.

The student will design and build these steps in a software component that integrates the data reliably. The student will include a diagram with the design and the components used next to the script. Python is recommended but not mandatory.

The result of the execution will be a database entity that can be queried and is compliant with the data dictionary of the dataset. If the schemas of the files do not match in one target system, the student might consider using polyglot persistence with different database flavors.

The student will validate that the data has been ingested correctly with simple queries that might be used downstream.

Statement

Brief of the assignment

- Motivation for the data topic and realm –expected insights
- The dataset as such -analysis
- The data architecture design
- The data ingestion pipeline
- Resulting data structures - validated
- Conclusions and further work

Deliverables

- Datasets and databases in native format (formats that can be used directly by means of SSMSs)
- 20-page long report explaining each required issue
- Set of diagrams, pictures, graphs, and graphical information in general to depict the adopted solution
- Source code of each batch process, query, script, or programmed used throughout the process

Delivery date

- 2nd of March 2025, 17:00.